

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTrainingData/Static · 425 trials
Total training paths	1 · 425 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/StaticReports

Sensor selection

Hands	left, right
Segments	thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	164 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	none
Feature mode	fft → feature dim 7544
Test size · seed · CV folds	0.2 · random_state=42 · 5-fold stratified
Sample counts	train (post-aug)=340 · test=85

Augmentation (training only)

Enabled	no
Copies per train trial	4
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (5-fold)	CV std	Hold-out test acc
SVM (Linear)	0.9971	0.0059	1.0000
Logistic Regression	0.9853	0.0093	1.0000
LDA	0.9647	0.0150	1.0000
KNN	0.9471	0.0220	0.9882
Random Forest	0.9647	0.0199	0.9765
SVM (RBF)	0.8618	0.0401	0.8353

Per-class hold-out F1

Class	SVM (Linear)	Logistic Regression	LDA	KNN	Random Forest	SVM (RBF)	Support
Double_Closed_Fist	1.000	1.000	1.000	1.000	0.963	0.765	13
Double_Nothing	1.000	1.000	1.000	1.000	0.957	0.846	12
Double_Okay	1.000	1.000	1.000	1.000	1.000	0.667	12
Double_Open_Palm	1.000	1.000	1.000	1.000	1.000	0.909	12
Double_Phone	1.000	1.000	1.000	0.960	0.957	0.857	12
Double_Pistol	1.000	1.000	1.000	0.957	1.000	0.857	12
Double_Spiderman	1.000	1.000	1.000	1.000	0.960	0.960	12
macro avg	1.000	1.000	1.000	0.988	0.977	0.837	85

External test (NewTestData)

Algorithm	External test acc
Random Forest	0.8056
LDA	0.7500
SVM (RBF)	0.3056
SVM (Linear)	0.3056
KNN	0.3056
Logistic Regression	0.1667

Per-class external accuracy

Class	Random Forest	LDA	SVM (RBF)	SVM (Linear)	KNN	Logistic Regression	Support
Double_Closed_Fist	0.333	1.000	0.000	0.000	0.000	0.000	6
Double_Nothing	1.000	1.000	0.400	0.200	0.200	0.200	5
Double_Okay	0.400	0.000	0.000	0.000	0.000	0.000	5

Double_Open_Palm	1.000	1.000	0.000	0.000	0.000	0.000	5
Double_Phone	1.000	0.200	0.000	0.000	0.000	0.000	5
Double_Pistol	1.000	1.000	0.800	1.000	1.000	1.000	5
Double_Spiderman	1.000	1.000	1.000	1.000	1.000	0.000	5